INCH-POUND

MIL-F-63281A (AR)
AMENDMENT 1
18 May 1990

MILITARY SPECIFICATION

FUZE, ROCKET: M433 LESS BOOSTER AND BOOSTER LEAD

This Amendment forms a part of Military Specification MIL-F-63281A (AR), dated 14 April 1988, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1, Add the following after "2-R-376 - 2.75 IN:...":

"9324106 - Rocket Management Subsystem, Inventory - Deployment, MI38".

PAGE 4

3.3.5, Delete "9260888" and substitute "12551812".

3.4.1.1, Delete in its entirety.

PAGE 5

3.4.1.2.b, Line 4, before: "The time delay shall be the time interval..." add the following:

"After the setting charge has been applied, wait 7 sec and then short pins 2 and 4 together.".

PAGE 6

3.4.2, Delete in its entirety.

PAGE 7

3.6.1.1, Delete in its entirety.

3.6.1.2, Delete "5 pounds minimum" and substitute "2.5 pounds minimum".

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3.6.2, Delete "9260756" and substitute "12551818".

3.6.3, Delete "9260756" and substitute "12551818".

Delete TABLE II in its entirety and substitute the following:

"TABLE II

Position of S&A Dev.	Terminals	Requirements
Safe	#1 & #2 (charge switch)	30 Megohms Min @ 500 VDC
Safe	#1 & #4 (charge switch)	100 Milliohms Max
Armed	#1 & #4 (charge switch)	30 Megohms Min @ 500 VDC
Armed	#3 & #5 & Rotor	100 Milliohms Max".

PAGE 8

3.8.2, Delete in its entirety and substitute:

"3.8.2 Charge circuit continuity (fuze unarmed). The fuze unarmed shall be tested for continuity through its umbilical lead. The contractor shall design a charge circuit continuity test set that provides an output pulse of 40 mamps maximum for a period of 0.10 msec maximum. The accept/reject requirement of the umbilical cable and the charge circuit of the firing capacitor is:

a. Reject as an open circuit if the measured resistance is greater than the resistance of a 2.25 kohm resistance connected in series with the charge circuit of the firing capacitor.

b. Reject as a short if the measured resistance is less than the resistance of 220 ohm resistance connected in parallel with the charge circuit of the firing capacitor.

The test set shall be calibrated with acceptable production modules using the above modifications. Thirty fuzes shall be modified for the short condition and thirty fuzes shall be modified for the open condition. The results of the calibration process will be subject to government review and approval."

PAGE 9

3.8.5.3, Add to the end of paragraph "To satisfy the requirement of MIL-STD-331, the disassembled fuze must display the following conditions:

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a. No loose explosive material shall be present in the fuze body assembly.

b. The lock roller pin on the rotor shall not be broken.

c. The rotor shall be in the safe position, secured such that it cannot be turned to the armed position with finger pressure.

d. The g-weight shall be in the fully safe position.

e. The switch assembly and its components shall not be extended, broken or loose."

3.8.5.4, Add to end of paragraph:

"e. The switch assembly and its components shall not be extended, broken or loose."

PAGE 10

3.8.5.5, Add to the end of paragraph "To satisfy the safety requirement, the disassembled fuze must display the following conditions:

"a. No loose explosive material shall be present in the fuze body assembly.

b. The lock roller pin on the rotor shall not be broken.

c. The rotor shall be in the safe position, secured such that it cannot be turned to the armed position with finger pressure.

d. The g-weight shall be in the fully safe position.

e. The switch assembly and its components shall not be extended, broken or loose."

PAGE 12

4.3.1, Delete "9332604" and substitute " "12551818".

PAGE 15

Table III: Delete second "Gear and pinion assembly No. 1" and substitute "Gear and pinion assembly No. 2".

PAGE 18

Table III:

Delete "9260752" and substitute "12551817".

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Delete "Pull-off test of switch cap 25 3.6.1.1 4.5.32".

PAGE 19

Table III: Delete "9332604" and substitute "12551818".

PAGE 21

Table III: Five Foot Drop Test, delete "25" and substitute "24".

Temperature and humidity test, delete "21" and substitute "22".

Ballistic firing test, delete "138" and substitute "240".

PAGE 25

4.4.2.1, Add Majors 118, 119 and 120 as follows:

"118	Wrong lubricant	100%	3.3	Certification
119	Excessive lubricant (a)	0.40%	3.3	Visual
120	Lubricant in not-called-for area	0.40%	3.3	Visual".

Add note to note block as follows:

"(a) Excessive lubricant is defined as evidence of non-uniform coating and collection of fluid in visible areas."

PAGE 29

4.4.2.3, Delete "9260750" and substitute "9260751".

PAGE 31

4.4.2.5, Delete "9260774" and substitute "9260751".

PAGE 37

4.4.2.11, Delete "9260751" and substitute "9260774".

PAGE 38

4.4.2.12, Delete "and Gear".

PAGE 46

4.4.2.24, Add majors 110 and 111:

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"110	Width of setback weight actuator pin clearance (.110-.005)	0.40%	3.3	Gage
111	Height of setback weight actuator pin clearance (.420+.010)	0.40%	3.3.	Gage".

PAGE 50

4.4.2.22, Add Critical 2 as follows:

"2	Pull-off test of switch assembly (a)	100%	3.6.1.2	4.5.33".

Delete Majors 102 and 103 in their entirety.

Delete note a in its entirety and substitute:

"(a) Perform Critical 2 before Critical 1."

PAGE 51

4.4.2.23, Delete "9332604" and substitute "12551818".

PAGE 52

4.4.2.24, Delete "9332604" and substitute "12551818".

PAGE 53

4.4.2.25:

Delete "9332603" and substitute "12551812".

Delete "9332604" and substitute "12551818".

PAGE 56

4.4.2.28, Add Majors 104, 105, 106, 107 and 108 as follows:

"104	Wrong adhesive	100%	3.3	Certification
105	Excessive adhesive (a)	0.40%	3.3	Visual
106	Adhesive in not-called-for area	0.40%	3.3	Visual
107	Improper care (time & condition) of adhesive	0.40%	3.3	Visual/Chart
108	Improper bonding of contact	0.40%	3.3	Visual".

Delete Minor 201 in its entirety.

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Add note to notes block as follows:

"(a) Adhesive is permitted in the immediate area around the contact/module interface as long as it does not extend into the contact cavity, make contact with the solder, or cause interference fit with the next assembly."

PAGE 62

4.4.2.32, Delete "9254707" and substitute "9260886".

Delete Major 101 in its entirety.

PAGE 64

4.4.2.33, Delete note c in its entirety.

PAGE 65

4.4.2.34, Add Majors 109, 110, 111, 112, 113, 114, 115 and 116 as follows:

"109	Wrong umbilical adhesive	100%	3.3	Certification
110	Umbilical adhesive missing	0.40%	3.3	Visual
111	Umbilical adhesive excessive (a)	0.40%	3.3	Visual
112	Umbilical adhesive in not-called-for area	0.40%	3.3	Visual
113	Wrong impact probe sealant	100%	3.3	Certification
114	Impact probe sealant missing	0.40%	3.3	Visual
115	Impact probe sealant excessive (b)	0.40%	3.3	Visual
116	Impact probe sealant in not-called-for area	0.40%	3.3	Visual".

Delete Minors 201 and 202 in their entirety.

Add notes to note block as follows:

"(a) Evidence of adhesive inside cavity of fuze body or failure to meet note 6 shall constitute excessive adhesive.

(b) Sealant is permitted in the immediate area around the joint of the probe and fuze body as long as it does not extend radically inward more than 0.10 inch or above the crimp lip."

PAGE 68

4.4.2.37:

Delete "9332603" and substitute "12551812".

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Delete Major 101 in its entirety.

PAGE 69

4.4.2.38, Delete "9332603" and substitute "12551812".

PAGE 70

4.4.2.39, Delete "9332603" and substitute "12551812".

PAGE 79

4.5.16, 7th line delete "three (3) seconds" and substitute "five (5) seconds".

PAGE 84

Add "4.5.24.1 Correlation procedure." before "For the correlation procedure,...".

PAGE 86

4.5.32, Delete in its entirety.

4.5.33, Delete in its entirety and substitute:

"4.5.33 Pull-off test of switch assembly. Secure the setback weight in a holding fixture. An axial load of 2.5 pounds minimum shall be applied to the switch cap. The total height of the weight, pin, insulator and cap assembly shall not increase by more than .001 in. due to this pull test. Any unit that fails this requirement shall be classified as defective."

PAGE 89

5.2, Delete "Dwg. 8886458" and substitute "Dwg. 8861213".

PAGE 91

Table IV: Delete "336" and substitute "366".

PAGE 92

Table IV:

Delete paragraph 3 in its entirety.

Delete paragraph 4 in its entirety.

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PAGE 93

Table IV:

Delete paragraph 6 in its entirety.

Paragraph 8, delete "Both setters shall be used.".

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